Results: damage_front

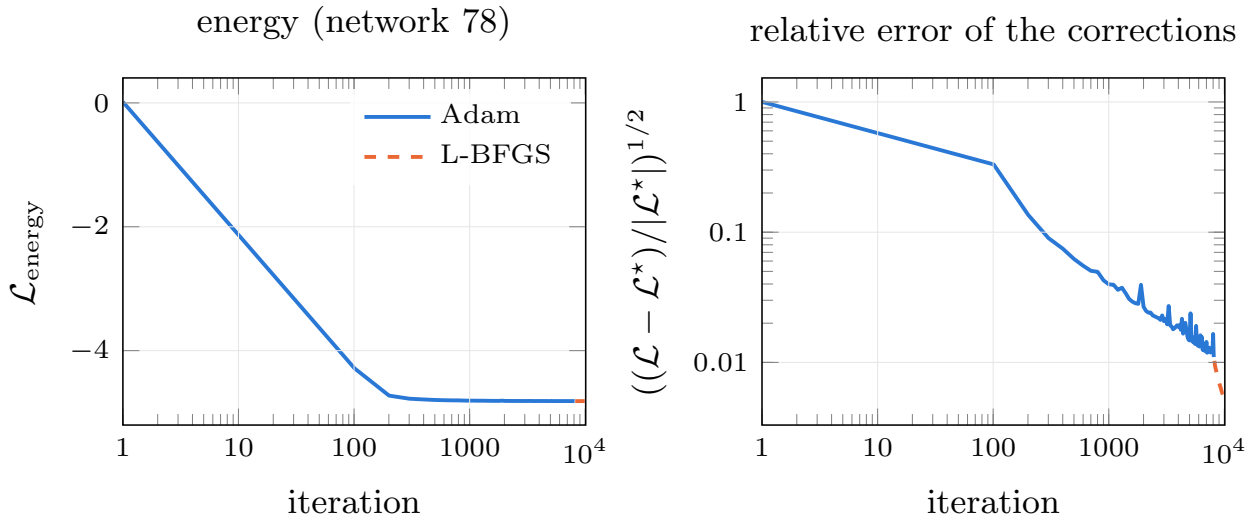


Figure 1: Training of the network 78: energy loss and relative error of the network corrections in the energy norm (Corollary 3.10) (`loss`)

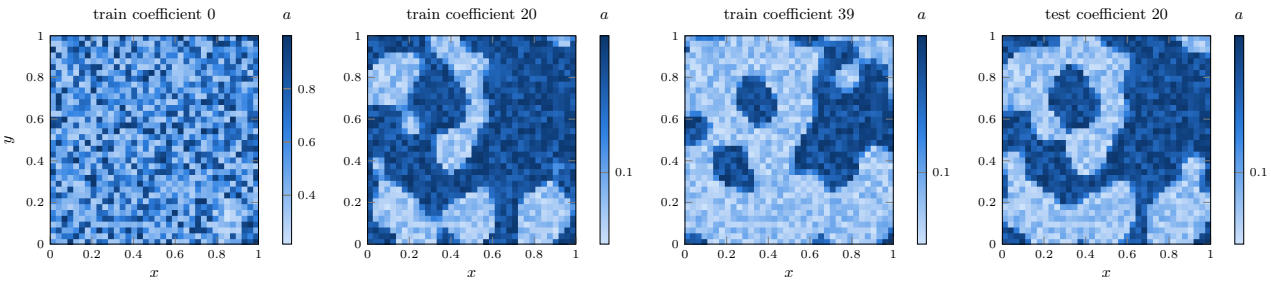


Figure 2: Coefficients of the train set and of the test set (`coefficients`)

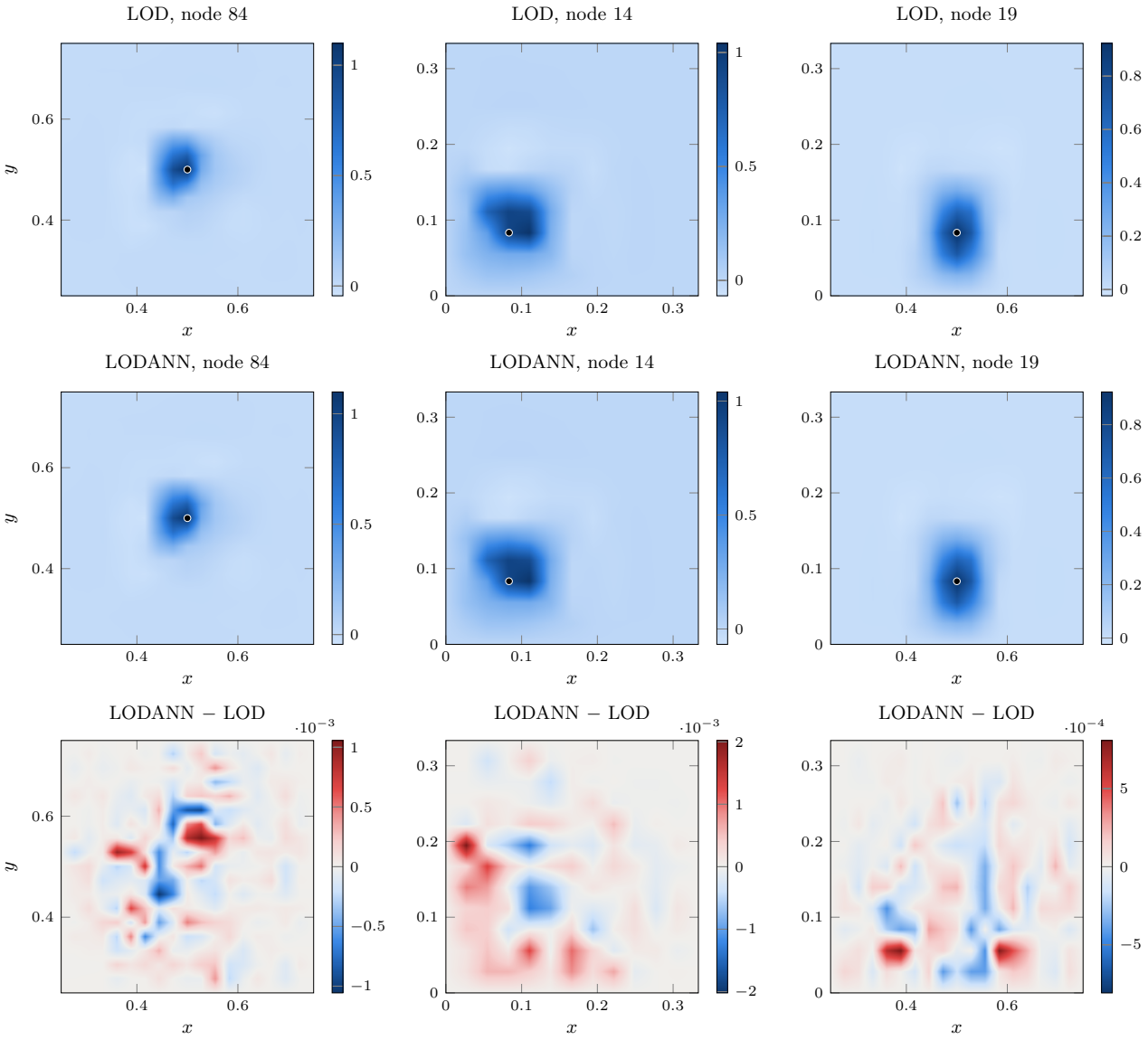


Figure 3: Corrected basis functions of LOD and LODANN and their difference (test coefficient 20) (`basis_functions`)

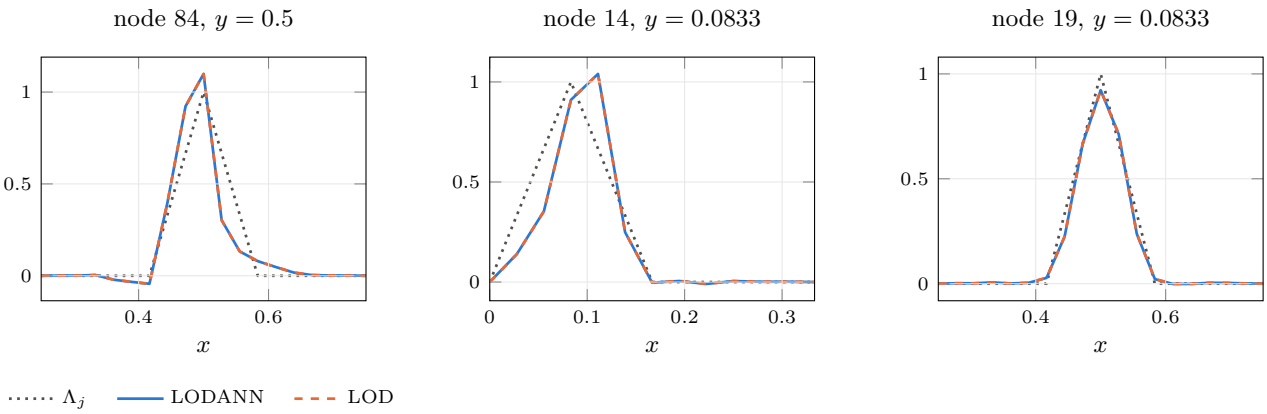


Figure 4: Cross sections of the basis functions through the node (`basis_cross_sections`)

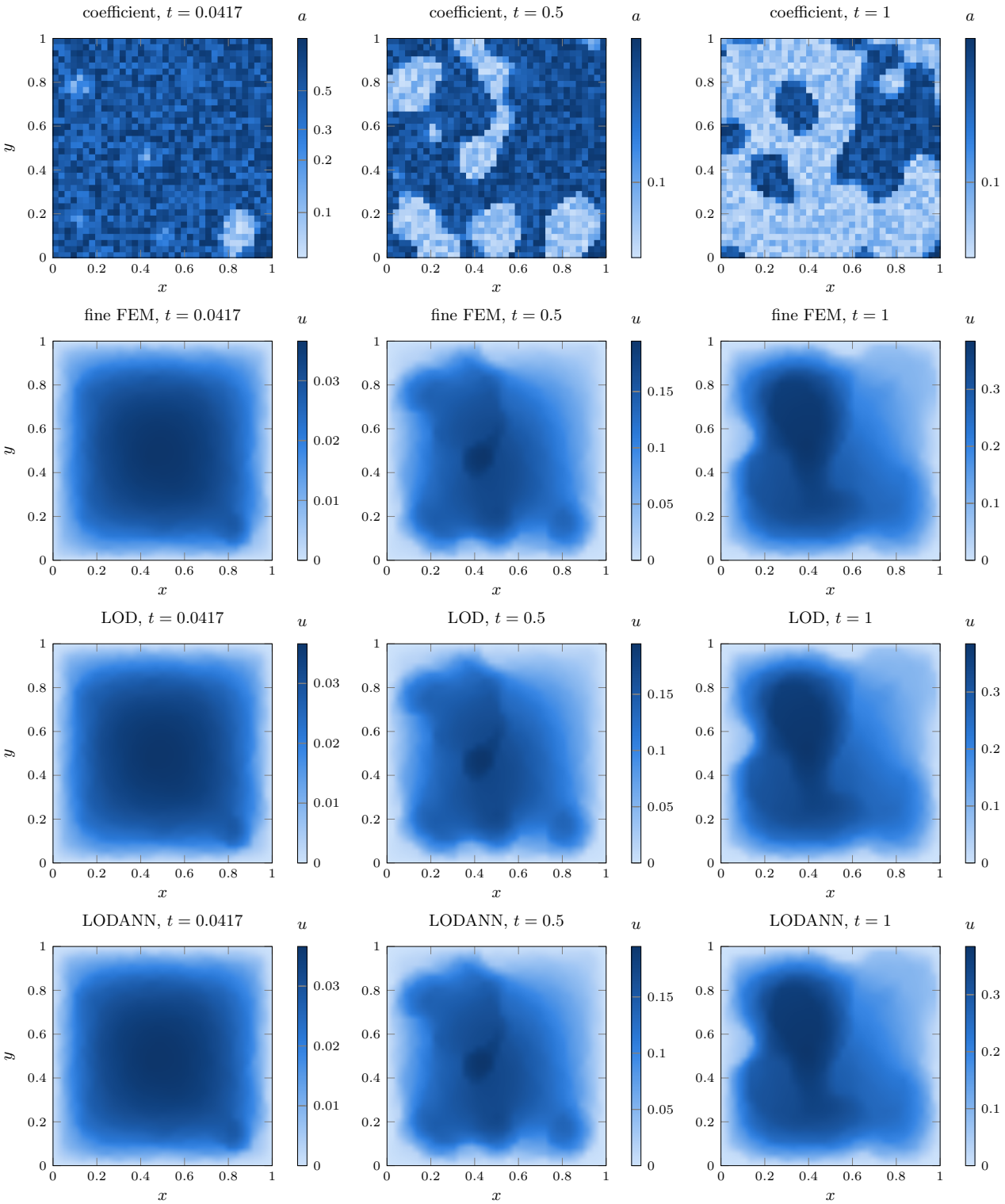


Figure 5: Coefficient, fine FEM (reference), LOD and LODANN solution (test set) (`solutions`)

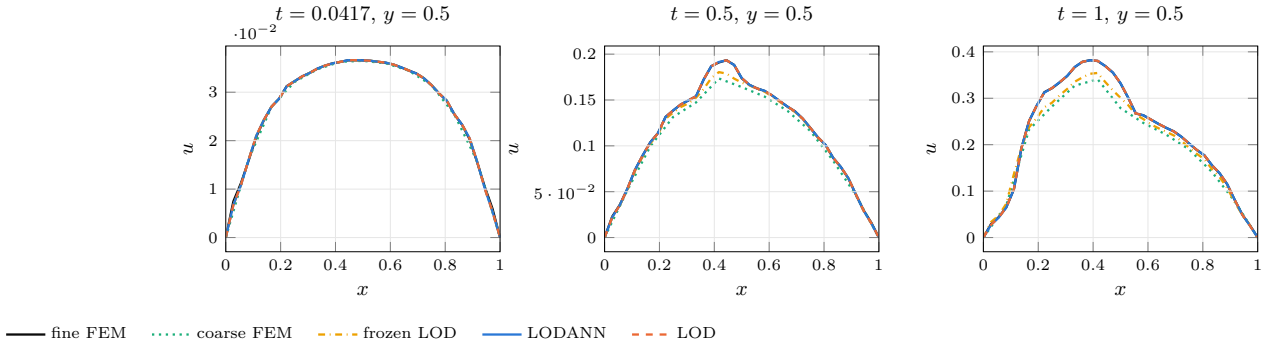


Figure 6: Cross sections of the solutions of all methods (test set) (`solution_cross_sections`)

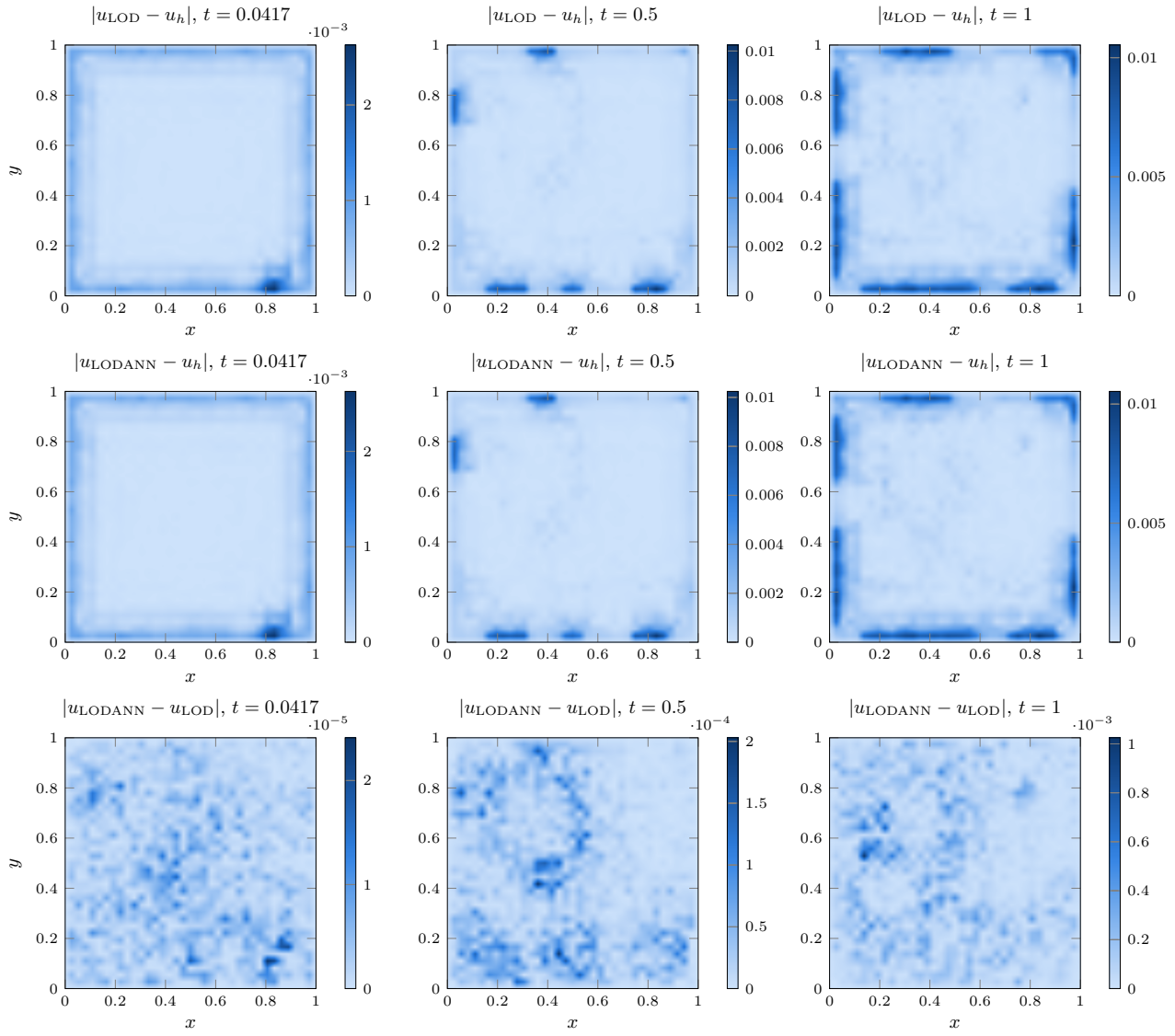


Figure 7: Pointwise errors against the fine FEM solution u_h and between LODANN and LOD (test set) (`pointwise_errors`)

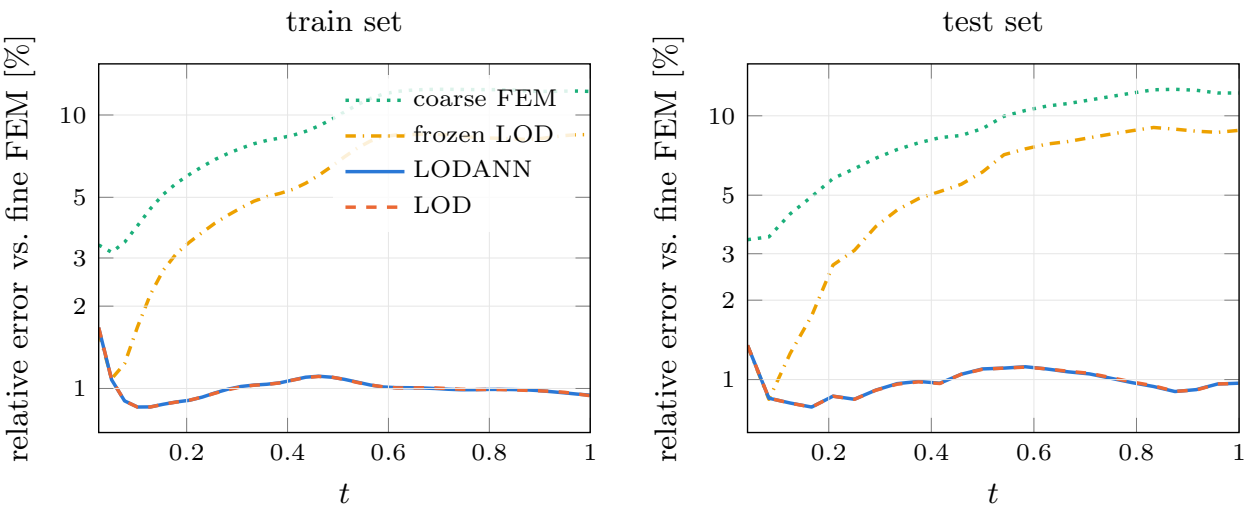


Figure 8: Relative error against the fine FEM solution over time (`errors_over_time`)

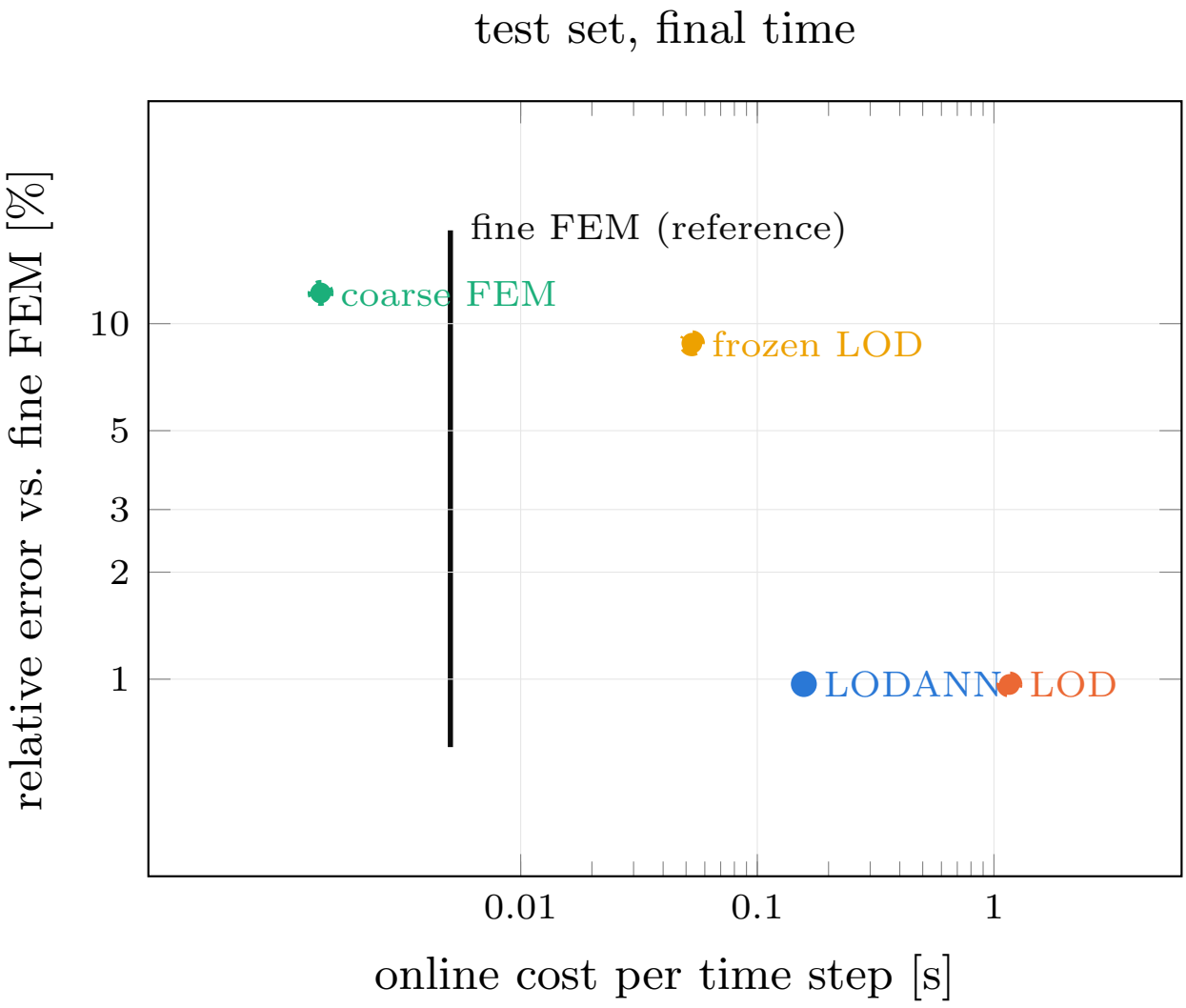


Figure 9: Accuracy against online cost per time step (test set). Fine FEM (reference): 0.00505 s per step; LODANN offline training: 5.62 h (recorded). The classical LOD timings refer to the unoptimized implementation in LOD.py. (`accuracy_cost`)

method	train: final	train: mean	test: final	test: mean	cost/step [s]
coarse FEM	12.19 %	9.36 %	12.20 %	9.00 %	0.00143
frozen LOD	8.47 %	6.07 %	8.81 %	5.87 %	0.0529
LOD	0.94 %	1.01 %	0.97 %	0.98 %	1.16
LODANN	0.95 %	1.01 %	0.97 %	0.98 %	0.157
fine FEM (reference)	–	–	–	–	0.00505
LODANN basis vs. LOD	0.16 %	–	0.15 %	–	

Figure 10: Relative errors against the fine FEM solution (final time and mean over time) and online cost per time step (`summary_table`)